

## DUNHUA, China

WMO: 541860

Lat: 43.3676N

Lon: 128.2110E

Elev: 1722

StdP: 13.80

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.7      | -10.9 | -23.0                         | 1.6 | -12.0 | -19.7 | 2.0 | -8.4 | 18.4                  | 6.8  | 16.7 | 8.9  | 2.5                   | 250  | 0.514 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.5                   | 85.2            | 68.6 | 82.6 | 67.7 | 80.1 | 66.6 | 72.9                | 80.9 | 71.3 | 78.3 | 69.7 | 76.3 | 6.6                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 70.4                            | 119.2 | 77.3 | 68.9 | 113.3 | 75.8 | 67.4 | 107.3 | 74.2 | 37.9          | 81.1 | 36.4 | 78.5 | 35.0 | 76.4 | 78.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               |      | Max  |                            | Min   | Max  | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 17.2              | 14.9 | 12.9 | DB                         | -18.6 | 89.8 | 4.9                | 2.9                                                | -22.1     | 91.9 | -25.0      | 93.6 | -27.7      | 95.2 | -31.3      | 97.3 |
|                   |      |      | WB                         | -18.9 | 74.8 | 4.7                | 1.8                                                | -22.3     | 76.2 | -25.0      | 77.2 | -27.7      | 78.3 | -31.0      | 79.6 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 40.0  | 4.8   | 12.2  | 26.1  | 42.6  | 55.0  | 64.7  | 70.2  | 68.2  | 57.5  | 43.0  | 24.7  | 9.3  | (6)  |
| (7)  |                                                                   | DBStd       | 24.18 | 7.88  | 9.62  | 10.08 | 8.95  | 7.12  | 5.79  | 4.85  | 5.04  | 6.16  | 8.08  | 10.78 | 8.99 | (7)  |
| (8)  |                                                                   | HDD50       | 5763  | 1401  | 1059  | 741   | 256   | 30    | 1     | 0     | 11    | 243   | 759   | 1262  |      | (8)  |
| (9)  |                                                                   | HDD65       | 9506  | 1866  | 1479  | 1205  | 673   | 319   | 74    | 13    | 25    | 234   | 682   | 1209  | 1727 | (9)  |
| (10) |                                                                   | CDD50       | 2115  | 0     | 0     | 1     | 35    | 185   | 441   | 626   | 565   | 236   | 26    | 0     | 0    | (10) |
| (11) |                                                                   | CDD65       | 382   | 0     | 0     | 0     | 1     | 9     | 64    | 174   | 125   | 9     | 0     | 0     | 0    | (11) |
| (12) |                                                                   | CDH74       | 2700  | 0     | 0     | 0     | 18    | 177   | 598   | 1143  | 689   | 73    | 2     | 0     | 0    | (12) |
| (13) |                                                                   | CDH80       | 574   | 0     | 0     | 0     | 1     | 32    | 136   | 274   | 128   | 3     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 5.3   | 4.9   | 5.5   | 6.6   | 6.7   | 6.1   | 4.9   | 4.6   | 4.2   | 4.2   | 5.1   | 5.4   | 4.9  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 25.0  | 0.2   | 0.2   | 0.4   | 1.3   | 2.4   | 4.1   | 6.1   | 6.0   | 2.4   | 1.1   | 0.6   | 0.2  | (15) |
| (16) |                                                                   | PrecMax     | 36.0  | 0.6   | 0.9   | 0.9   | 2.8   | 4.4   | 8.2   | 11.2  | 12.8  | 6.0   | 2.4   | 1.4   | 1.5  | (16) |
| (17) |                                                                   | PrecMin     | 14.1  | 0.0   | 0.0   | 0.0   | 0.4   | 0.4   | 0.8   | 1.5   | 1.8   | 0.5   | 0.1   | 0.1   | 0.0  | (17) |
| (18) |                                                                   | PrecStd     | 4.8   | 0.1   | 0.2   | 0.2   | 0.6   | 1.1   | 2.0   | 2.2   | 2.8   | 1.5   | 0.7   | 0.4   | 0.3  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 31.6  | 42.5  | 59.1  | 76.9  | 84.1  | 87.7  | 89.8  | 87.1  | 79.6  | 71.7  | 56.4  | 38.6 | (19) |
| (20) |                                                                   | 0.4%        | MCWB  | 25.9  | 33.0  | 44.1  | 53.7  | 59.7  | 64.7  | 71.7  | 72.2  | 64.7  | 54.9  | 44.9  | 33.6 | (20) |
| (21) |                                                                   | 2%          | DB    | 26.1  | 37.4  | 51.1  | 69.7  | 79.1  | 83.8  | 85.7  | 83.3  | 76.0  | 66.2  | 50.1  | 33.2 | (21) |
| (22) |                                                                   | 2%          | MCWB  | 22.2  | 30.3  | 38.8  | 50.3  | 57.1  | 64.3  | 70.7  | 70.7  | 62.1  | 51.7  | 41.1  | 28.8 | (22) |
| (23) |                                                                   | 5%          | DB    | 22.3  | 33.4  | 46.2  | 64.5  | 74.6  | 80.6  | 83.0  | 80.5  | 72.8  | 61.4  | 45.3  | 28.7 | (23) |
| (24) |                                                                   | 5%          | MCWB  | 18.9  | 27.9  | 36.0  | 47.6  | 55.8  | 63.4  | 69.8  | 69.0  | 59.8  | 49.3  | 38.3  | 25.0 | (24) |
| (25) |                                                                   | 10%         | DB    | 18.4  | 29.2  | 41.7  | 59.4  | 70.0  | 77.2  | 80.3  | 77.8  | 69.9  | 57.3  | 41.0  | 24.3 | (25) |
| (26) |                                                                   | 10%         | MCWB  | 15.5  | 24.4  | 33.3  | 44.7  | 53.7  | 62.9  | 69.0  | 67.9  | 58.4  | 47.1  | 35.3  | 21.3 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 26.8  | 34.2  | 45.1  | 57.7  | 63.1  | 69.3  | 75.6  | 74.5  | 68.0  | 56.6  | 46.0  | 34.3 | (27) |
| (28) |                                                                   | 0.4%        | MCDWB | 30.5  | 39.6  | 57.6  | 71.4  | 77.8  | 79.7  | 84.7  | 83.7  | 74.5  | 68.6  | 53.7  | 37.9 | (28) |
| (29) |                                                                   | 2%          | WB    | 22.4  | 31.1  | 40.0  | 51.9  | 59.7  | 67.5  | 73.6  | 72.7  | 65.2  | 53.6  | 42.1  | 29.3 | (29) |
| (30) |                                                                   | 2%          | MCDWB | 25.8  | 36.8  | 49.3  | 68.8  | 73.1  | 77.4  | 81.9  | 80.5  | 71.8  | 63.4  | 48.9  | 32.9 | (30) |
| (31) |                                                                   | 5%          | WB    | 19.0  | 28.0  | 36.6  | 48.4  | 57.4  | 66.0  | 72.0  | 71.2  | 62.7  | 51.1  | 39.0  | 25.0 | (31) |
| (32) |                                                                   | 5%          | MCDWB | 22.3  | 33.5  | 45.0  | 62.2  | 70.4  | 76.2  | 79.4  | 77.8  | 69.6  | 59.0  | 44.7  | 28.4 | (32) |
| (33) |                                                                   | 10%         | WB    | 15.4  | 24.5  | 33.7  | 45.6  | 55.2  | 64.4  | 70.4  | 69.8  | 60.5  | 48.5  | 35.4  | 21.3 | (33) |
| (34) |                                                                   | 10%         | MCDWB | 18.2  | 29.1  | 41.0  | 57.8  | 68.0  | 73.9  | 77.6  | 75.4  | 67.5  | 55.9  | 40.6  | 24.4 | (34) |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 19.8  | 21.0  | 19.0  | 20.5  | 20.7  | 18.7  | 15.5  | 15.8  | 21.4  | 21.6  | 17.5  | 17.9 | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 24.0  | 24.5  | 23.9  | 29.5  | 29.7  | 25.1  | 20.6  | 19.4  | 25.3  | 28.7  | 23.2  | 21.6 | (36) |
| (37) |                                                                   |             | MCWBR | 20.5  | 18.5  | 14.1  | 14.5  | 12.0  | 9.3   | 7.9   | 8.2   | 12.1  | 15.7  | 15.4  | 18.1 | (37) |
| (38) |                                                                   |             | MCDBR | 23.7  | 23.2  | 22.2  | 26.8  | 24.8  | 19.9  | 16.4  | 16.1  | 18.9  | 25.1  | 21.5  | 21.0 | (38) |
| (39) |                                                                   | 5% WB       | MCWBR | 20.5  | 17.8  | 14.0  | 14.0  | 11.1  | 8.8   | 7.4   | 7.8   | 10.1  | 14.9  | 15.1  | 18.0 | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.333 | 0.380 | 0.457 | 0.542 | 0.584 | 0.589 | 0.553 | 0.491 | 0.438 | 0.464 | 0.397 | 0.347 | (40) |      |
| (41) |                                                                   | taud        | 2.038 | 1.935 | 1.823 | 1.724 | 1.689 | 1.756 | 1.896 | 2.056 | 2.113 | 1.919 | 1.986 | 2.018 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 239   | 246   | 242   | 232   | 227   | 226   | 233   | 244   | 247   | 215   | 210   | 218   | (42) |      |
| (43) |                                                                   | Edh at Noon | 35    | 46    | 59    | 70    | 75    | 70    | 61    | 50    | 43    | 47    | 37    | 32    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 743   | 1068  | 1394  | 1491  | 1651  | 1705  | 1534  | 1486  | 1369  | 985   | 660   | 584   | (44) |      |
| (45) |                                                                   | RadStd      | 55    | 56    | 88    | 105   | 143   | 173   | 158   | 136   | 107   | 79    | 63    | 41    | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)  | (e)                    | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) | Station Only           |       | N/A     | N/A    | -1.73   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (4 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page